

Pandrosion, Thales Scaling, and a Steffensen Acceleration

A readable account of a geometric scheme for the equation $z^p = x$

Ivan BESEVIC
Independent Mathematical Research
github.com/ivan-fr/poussiere_de_x

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Abstract

Pandrosion of Alexandria is associated with a geometric approximation to the problem of inserting proportional means, and hence to the classical problem of extracting cube roots. This paper rewrites that construction in modern notation and follows the consequences as far as possible while keeping the geometry visible.

The central object is a derivative-free fixed-point map for the equation $z^p = x$. On the real line it takes the universal form

$$h_{p,x}(s) = 1 - \frac{x-1}{x(1+s+\cdots+s^{p-1})}.$$

Its fixed point is $s_* = x^{-1/p}$, and the desired root is recovered by $v = xs_*^{p-1} = x^{1/p}$. The contraction factor is explicit:

$$\lambda_{p,x} = \frac{(\alpha-1) \sum_{k=1}^{p-1} k \alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k}, \quad \alpha = x^{1/p}.$$

This expression explains why large targets slow the raw construction and why homothetic preconditioning, or “Thales scaling”, can dramatically reduce the number of parallel constructions needed.

The map is then accelerated by Aitken–Steffensen’s Δ^2 transform, which turns the linearly convergent Pandrosion iterator into a quadratically convergent derivative-free scheme. The complex extension is obtained by working near the p cyclotomic anchors of x^{-1} and by seeding points into local contraction disks. The final section explains how the same fixed-point idea extends from monomials to general monic polynomials.

The paper is intentionally modest in scope: it does not claim a universal advance over modern polynomial solvers. Its purpose is to present, in a clear and self-contained form, the mathematics behind a concrete geometric root-finding scheme, its residual profile, its scaling principle, and its complex multi-start extension.

***One-sentence summary.** Pandrosion supplies a geometric fixed-point map; Steffensen accelerates it; homothety makes the geometry smaller before the map starts working.*

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1 A reader's guide

The paper is organized around one question:

What does Pandrosion's ruler-and-parallels construction become when it is written as an iteration?

The answer is surprisingly compact. A single rational map appears, its fixed point is explicit, and its convergence factor can be computed in closed form. The rest of the paper refines this basic observation.

- Sections 2–4 derive the iteration from the cube-root case and then state the general p -th root form.
- Sections 5–6 analyze the residual profile: the contraction factor, monotonicity, and error bounds.
- Sections 8–10 discuss the practical improvements: a better first point, Steffensen acceleration, and scaling.
- Sections 12–14 explain the complex multi-start picture and the formal-verification layer described in the original development.
- Section 15 explains what remains true for a general monic polynomial.

The exposition is deliberately less compressed than the original manuscript. The goal is not to hide the algebra, but to make clear what each formula is for.

2 The cube-root construction

Pandrosion's construction begins with the ancient problem of doubling the cube: constructing a segment of length $\sqrt[3]{2}$ from a unit segment. Algebraically, this is the problem of inserting two proportional means m_1, m_2 between 1 and 2:

$$\frac{1}{m_1} = \frac{m_1}{m_2} = \frac{m_2}{2}.$$

The solution is $m_1 = \sqrt[3]{2}$ and $m_2 = \sqrt[3]{4}$.

Pandrosion's geometric procedure does not produce the exact value in finitely many steps. It produces a sequence of approximations. This is the key modern viewpoint: the construction is an iteration, and the error sequence of that iteration is its residual profile.

In the classical cube-root case one may write the construction with an internal length u_n and a visible readout v_n :

$$u_{n+1} = \frac{u_n}{2 - 2\left(\frac{4-u_n}{4}\right)^3}, \quad v_n = 2\left(\frac{4-u_n}{4}\right)^2. \quad (1)$$

The number u_n belongs to the diagram. The number v_n is the approximation that is read from the diagram.

The substitution

$$s_n = \frac{4 - u_n}{4}$$

removes the distracting constants. Since $u_n = 4(1 - s_n)$, the readout becomes $v_n = 2s_n^2$, and the iteration becomes

$$s_{n+1} = \frac{2s_n^2 + 2s_n + 1}{2s_n^2 + 2s_n + 2} = 1 - \frac{1}{2(1 + s_n + s_n^2)}. \quad (2)$$

Proposition 2.1 (Cube-root fixed point). *The map in (2) has the fixed point*

$$s_* = 2^{-1/3}.$$

Consequently the readout satisfies

$$v_n = 2s_n^2 \longrightarrow 2s_*^2 = \sqrt[3]{2}.$$

Proof. The fixed-point equation is

$$s = 1 - \frac{1}{2(1 + s + s^2)}.$$

Multiplying by $2(1 + s + s^2)$ gives

$$2s(1 + s + s^2) = 2(1 + s + s^2) - 1,$$

which simplifies to $2s^3 = 1$. Hence $s = 2^{-1/3}$ in the positive real branch. The readout identity follows immediately. $\square$

The diagram computes $\sqrt[3]{2}$ indirectly. It first converges to $2^{-1/3}$, and the visible root is recovered by the readout $v = 2s^2$.

3 Why the polynomial S_p appears

The polynomial

$$S_p(s) = 1 + s + \cdots + s^{p-1}$$

appears for a geometric reason. Thales' theorem constructs proportional segments. Repeating the construction builds a finite geometric progression of ratios. Algebraically that progression is exactly S_p .

The elementary identity behind the construction is

$$1 - s^p = (1 - s)(1 + s + \cdots + s^{p-1}) = (1 - s)S_p(s). \quad (3)$$

Thus, whenever a diagram compares 1 and s^p through successive proportional segments, the denominator $S_p(s)$ is the natural object that records the accumulated proportional means.

This identity is the bridge between the geometric construction and the fixed-point map.

4 The universal real-line iteration

Let $p \geq 2$ and $x > 0$. Define

$$S_p(s) = \sum_{k=0}^{p-1} s^k, \quad h_{p,x}(s) = 1 - \frac{x - 1}{xS_p(s)}. \quad (4)$$

The Pandrosion iteration is

$$s_{n+1} = h_{p,x}(s_n). \quad (5)$$

The desired root is not s_n itself. It is read from s_n by

$$v_n = xs_n^{p-1}. \quad (6)$$

Theorem 4.1 (Fixed point and readout). *Let $x > 1$ and $p \geq 2$. The map $h_{p,x}$ has the positive fixed point*

$$s_* = x^{-1/p}.$$

At this fixed point the readout is

$$xs_*^{p-1} = x^{1/p}.$$

Proof. The fixed-point equation $s = h_{p,x}(s)$ is

$$1 - s = \frac{x - 1}{xS_p(s)}.$$

Using (3), this is equivalent to

$$x(1 - s)S_p(s) = x - 1, \quad \text{hence} \quad x(1 - s^p) = x - 1.$$

Therefore $xs^p = 1$, so $s = x^{-1/p}$ in the positive branch. The readout is then $x(x^{-1/p})^{p-1} = x^{1/p}$. $\square$

Remark 4.2. The map $h_{p,x}$ is derivative-free in its definition. It is built from the finite geometric sum S_p , not from the derivative of $s^p - x$.

5 The contraction ratio

The speed of the raw Pandrosion iteration is governed by the derivative of $h_{p,x}$ at the fixed point. This derivative is not used to define the method; it is used only to analyze the residual profile.

Let

$$\alpha = x^{1/p}.$$

Then $s_* = \alpha^{-1}$.

Theorem 5.1 (Closed-form contraction factor). *For $x > 1$ and $p \geq 2$, the linear contraction factor of the Pandrosion iteration at the fixed point is*

$$\lambda_{p,x} = h'_{p,x}(s_*) = \frac{(\alpha - 1) \sum_{k=1}^{p-1} k \alpha^{p-1-k}}{\sum_{k=0}^{p-1} \alpha^k}. \quad (7)$$

In particular,

$$\lambda_{2,x} = \frac{\alpha - 1}{\alpha + 1}, \quad \lambda_{3,x} = \frac{(\alpha - 1)(\alpha + 2)}{\alpha^2 + \alpha + 1}.$$

Proof. Differentiate (4):

$$h'_{p,x}(s) = \frac{x - 1}{x} \frac{S'_p(s)}{S_p(s)^2}.$$

Substitute $s = \alpha^{-1}$ and use

$$S_p(\alpha^{-1}) = \sum_{k=0}^{p-1} \alpha^{-k} = \alpha^{1-p} \sum_{k=0}^{p-1} \alpha^k.$$

The same change of powers in $S'_p(\alpha^{-1})$ yields (7). $\square$

The formula immediately explains the qualitative behavior:

- if x is close to 1, then $\lambda_{p,x}$ is small and the raw geometry converges quickly;
- if x is large, then $\lambda_{p,x}$ approaches 1 and the raw geometry converges slowly;
- increasing p at fixed large target makes the unscaled construction more delicate.

Proposition 5.2 (Large- p limit at fixed target). *For fixed $x > 1$,*

$$\lim_{p \rightarrow \infty} \lambda_{p,x} = 1 - \frac{\log x}{x - 1}.$$

Proof sketch. Write $\alpha = x^{1/p} = e^{(\log x)/p}$ and replace the sums in (7) by Riemann sums on $[0, 1]$. The resulting quotient reduces to the displayed expression. $\square$

The number $\lambda_{p,x}$ is the fingerprint of Pandrosion's residual profile. Scaling works because it changes x before this fingerprint is formed.

6 Error bounds and iteration counts

The local statement is the usual one for a contracting fixed-point map. If an interval I contains s_* and

$$\sup_{s \in I} |h'_{p,x}(s)| \leq \Lambda < 1,$$

then every orbit that remains in I satisfies

$$|s_n - s_*| \leq \Lambda^n |s_0 - s_*|. \quad (8)$$

Consequently the readout error is controlled by the same geometric decay, with an additional Lipschitz constant for $s \mapsto xs^{p-1}$ on I .

A practical iteration budget follows from (8). To reach $|s_n - s_*| \leq \tau$, it is enough to take

$$n \geq \frac{\log(\tau/|s_0 - s_*|)}{\log \Lambda}. \quad (9)$$

The denominator is negative, so a smaller Λ means a smaller budget. This is the numerical meaning of the geometric statement: fewer effective Thales parallels are needed.

7 The case $0 < x < 1$

When $0 < x < 1$, the fixed point $s_* = x^{-1/p}$ lies outside $(0, 1)$. The direct real-line diagram that was natural for $x > 1$ therefore no longer lives in the same interval. One can still analyze the rational map, but the clean geometric contraction picture changes.

The practical resolution is simple: use the reciprocal target. Since

$$x^{1/p} = \frac{1}{(1/x)^{1/p}},$$

one applies the $x > 1$ theory to $1/x$ and then inverts the result. This is the first appearance of a principle that becomes important in the complex theory: changing charts is often better than forcing one chart to handle every scale.

8 A better starting point

The raw iteration needs an initial point. The most natural choice is not always the fastest. A simple geometric improvement is to start from the image of the neutral point 1:

$$s_0^{\text{opt}} = h_{p,x}(1) = 1 - \frac{x-1}{xp}. \quad (10)$$

This point already incorporates one full proportional-mean step. In the language of the diagram, one avoids spending future parallels to do work that can be done once at initialization.

Proposition 8.1 (Canonical start). *The start $s_0^{\text{opt}} = h_{p,x}(1)$ is the canonical one-step start produced by the same Pandrosion geometry. Near $x = 1$ it satisfies*

$$s_0^{\text{opt}} = x^{-1/p} + O((x-1)^2).$$

Proof. The first statement is the definition. The expansion follows from $x^{-1/p} = 1 - (x-1)/p + O((x-1)^2)$ and from $1 - (x-1)/(xp) = 1 - (x-1)/p + O((x-1)^2)$. $\square$

Starting-point optimization is not an extra algorithmic layer. It is the first Pandrosion step used deliberately, before the iteration budget begins.

9 Steffensen acceleration

The raw Pandrosion map is geometrically meaningful but only linearly convergent. Aitken–Steffensen acceleration turns any sufficiently regular linearly convergent fixed-point iteration into a quadratically convergent one, without requiring derivatives.

For $h = h_{p,x}$, define

$$\sigma_{p,x}(s) = s - \frac{(h(s) - s)^2}{h(h(s)) - 2h(s) + s}. \quad (11)$$

Theorem 9.1 (Quadratic acceleration). *Assume h is analytic near s_* , $h(s_*) = s_*$, and $h'(s_*) \neq 1$. Then the Steffensen map $\sigma_{p,x}$ has the same fixed point and satisfies*

$$\sigma_{p,x}(s) - s_* = C(s - s_*)^2 + O((s - s_*)^3)$$

for an explicit constant C depending on the first derivatives of h at s_* . Thus the accelerated Pandrosion iteration is locally quadratic.

Proof sketch. Expand $h(s_* + e)$ as

$$s_* + \lambda e + ae^2 + O(e^3).$$

Substitute this expansion into (11). The linear term cancels, leaving a quadratic leading term. $\square$

Steffensen acceleration is especially natural here. The underlying geometric map remains derivative-free, and the acceleration uses only evaluations of the same map.

10 Thales scaling

Scaling is the most important practical improvement. Suppose x is large. Instead of applying Pandrosion directly to x , write

$$x = Ax', \quad (12)$$

where A is chosen so that x' is close to 1. Then

$$x^{1/p} = A^{1/p}(x')^{1/p}.$$

One computes $(x')^{1/p}$ by Pandrosion and multiplies back by $A^{1/p}$.

The point is not cosmetic. The contraction factor is controlled by x' , not by x . Since $\lambda_{p,x}$ becomes small when x is close to 1, the scaled problem may require far fewer effective Thales constructions.

Homothety does not merely normalize the input. It changes the geometry before Pandrosion begins. A good homothety reduces the number of proportional layers the construction must build.

This is the one-dimensional ancestor of the geometric scaling used later in multivariate Pandrosion solvers: first move the problem into a stable chart, then apply the local correction.

11 Comparison with standard methods

For $f(u) = u^p - x$, Newton's method is

$$u_{n+1} = u_n - \frac{u_n^p - x}{pu_n^{p-1}}.$$

It is quadratically convergent near a simple root, but it uses the derivative of f . Halley's method and the higher Schröder–König family can converge with even higher order.

Pandrosion is different in spirit. Its raw form is linearly convergent, but it arises from a proportional geometry and uses only the finite sum S_p . After Steffensen acceleration, it becomes derivative-free and quadratic. After scaling, it can be made numerically competitive in regimes where the geometric preconditioner makes x' close to 1.

Method	Local order	Uses derivative?
Raw Pandrosion	linear	no
Steffensen–Pandrosion	quadratic	no
Newton	quadratic	yes
Halley	cubic	yes

The intended message is not that Pandrosion dominates Newton. It usually does not. The message is that Pandrosion supplies a geometric fixed-point operator whose residual profile is exactly analyzable and whose performance improves substantially under scaling and Steffensen acceleration.

12 Complex extension

The real-line analysis identifies one positive fixed point. Over $\mathbb{C}$, the equation $s^p = 1/x$ has p fixed points:

$$\gamma_k = x^{-1/p} \zeta_p^k, \quad \zeta_p = e^{2\pi i/p}, \quad k = 0, \dots, p-1. \quad (13)$$

These are the cyclotomic anchors of the complex Pandrosion map.

The complex dynamics has non-trivial basin structure. A practical way to avoid this difficulty is to seed points directly into a known local contraction disk around the desired anchor.

Definition 12.1 (Targeted seed). Given an anchor γ and a parameter $0 < \varepsilon < 1$, define

$$\text{seed}_{\gamma,\varepsilon}(z) = \gamma + \varepsilon(z - \gamma).$$

The seed is a homothety centered at γ . It pulls any point z closer to the target anchor. For ε small enough, the seeded point lies in a local quadratic-convergence region for the Steffensen–Pandrosion map.

Theorem 12.2 (Seeded complex convergence). *Let γ be one of the anchors (13). There exists a radius $r_\gamma > 0$ such that if*

$$|\text{seed}_{\gamma,\varepsilon}(z) - \gamma| < r_\gamma,$$

then the Steffensen–Pandrosion orbit converges quadratically to γ . Moreover, choosing ε small enough enforces this condition for every fixed $z \in \mathbb{C}$.

Proof sketch. The Steffensen map is analytic near a simple fixed point and has a quadratic fixed point there. Hence a sufficiently small disk around γ is mapped into itself with quadratic contraction. The seed map shrinks the distance to γ by the factor ε . $\square$

Once γ_k is obtained, the corresponding root of x is reconstructed by

$$\rho_k = x \gamma_k^{p-1} = x^{1/p} \zeta_p^{-k}.$$

Thus the complex scheme is a multi-start method indexed by the cyclotomic anchors.

13 Measurability and continuity of the solution selector

A target selector chooses which anchor should receive a given starting point. The simplest selector is the nearest-anchor rule: assign z to the anchor γ_k that minimizes $|z - \gamma_k|$.

This selector is discontinuous on the Voronoi boundaries between anchors, but these boundaries have Lebesgue measure zero in the plane. Away from them, the chosen anchor is locally constant, and the seeded iteration depends continuously on the input.

Proposition 13.1 (A.e. continuity). *The solution map induced by the nearest-anchor selector and the seeded Steffensen–Pandrosion iteration is Borel measurable everywhere and continuous outside the Voronoi boundary of the anchor set.*

Proof sketch. Voronoi cells are defined by finitely many inequalities of continuous functions, hence are Borel. Their boundaries are finite unions of real codimension-one sets. Inside a cell, the same anchor is selected and the iteration is a locally uniform limit of continuous functions. $\square$

14 Formal verification layer

The original development reports a Lean 4 / Mathlib v4.7.0 corpus formalizing the elementary algebraic backbone of the construction. Its role is not to certify every numerical experiment. Its role is to pin down the identities on which the scheme rests:

- the finite-sum identity $(1 - s)S_p(s) = 1 - s^p$;
- the fixed-point equation for $h_{p,x}$;
- the reconstruction identity $xs_*^{p-1} = x^{1/p}$;
- the compatibility of seeding, target selection, and reconstruction in the complex multi-start setting.

The reported corpus contains 121 modules with no remaining `sorry` and with axioms restricted to the standard Lean whitelist described in the original project notes. This is best understood as a formal audit of the algebraic core, not as a claim that the numerical method is globally optimal.

15 Beyond monomials

The monomial equation $z^p = x$ is the clean case because the finite geometric sum S_p gives a closed-form denominator. Nevertheless the same fixed-point idea extends to any monic polynomial

$$P(z) = z^p + R(z).$$

Define

$$h_P(s) = 1 - \frac{R(s) + 1}{S_p(s)}. \tag{14}$$

Proposition 15.1 (Fixed points of the generalized Pandrosion map). *A point s is a fixed point of h_P if and only if it is a root of P .*

Proof. The equation $s = h_P(s)$ is

$$1 - s = \frac{R(s) + 1}{S_p(s)}.$$

Multiplying by $S_p(s)$ and using $(1-s)S_p(s) = 1 - s^p$ gives

$$1 - s^p = R(s) + 1,$$

which is equivalent to $s^p + R(s) = 0$. □

This extension preserves the derivative-free character of the fixed-point map, but it loses the clean closed-form contraction profile of the monomial case. The derivative $h'_p(\alpha)$ depends on the full polynomial R , and roots may be near singularities of the rational map. Therefore the monomial theory is not automatically a uniform theory for all polynomials.

The monomial case is special because geometry, algebra, and convergence all collapse to one finite sum. General polynomials keep the fixed-point idea but not the same closed-form speed guarantees.

16 Conclusion

Pandrosion's construction can be read as a root-finding algorithm. In modern notation, it is the fixed-point scheme

$$s_{n+1} = 1 - \frac{x - 1}{x(1 + s_n + \cdots + s_n^{p-1})}.$$

Its fixed point is $x^{-1/p}$, its readout is $x^{1/p}$, and its contraction ratio is the explicit quantity (7).

Three ideas make the method practically interesting:

1. **Starting-point optimization:** begin from a point already shaped by the Pandrosion geometry.
2. **Steffensen acceleration:** upgrade the derivative-free fixed point map from linear to quadratic convergence.
3. **Thales scaling:** use homothety to reduce the target before the geometric construction starts.

The complex extension adds cyclotomic anchors and seeded local convergence. The polynomial extension shows that the fixed-point idea is not restricted to monomials, although the monomial case is the one where the residual profile is fully explicit.

The main lesson is geometric: before asking an iteration to converge, one may change the space in which it works. In the Pandrosion viewpoint, homothety is not a numerical trick. It is part of the construction itself.

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